

Modulation of Myosin Heavy Chains in Rat Laryngeal Muscle

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Objectives: To test the hypothesis that myosin heavy chain (MHC) composition is a biological marker indicative of appropriate and functional reinnervation. **Study Design:** Age-matched adult rats were randomized for prospective study under three experimental conditions. **Methods:** In adult rats, three experimental conditions were surgically created, including transient recurrent laryngeal nerve (RLN) crush injury, RLN transection and repair, and cricoarytenoid joint fixation with intact RLN. Animals were survived for 30, 90, and 180 days. At each interval, vocal fold mobility was assessed by rigid microlaryngoscopy. Laryngeal electromyography (EMG) was performed before euthanasia. The thyroarytenoid and posterior cricoarytenoid muscles were then excised, each muscle was processed for sodium dodecyl sulfate–polyacrylamide gel electrophoresis, and MHC composition was determined. **Results:** Thirty days after nerve crush injury, three of six animals regained vocal fold mobility and normal MHC composition. Impaired vocal fold motion in three of six animals was associated with MHC composition characteristic of denervation. At 90 and 180 days, normal vocal fold motion and normal MHC composition were observed in all animals. Following nerve transection and repair, impaired vocal fold motion and MHC composition characteristic of denervation were observed in all animals, despite evidence of reinnervation on EMG. Following joint fixation, alteration in MHC composition consistent with denervation was observed only at 30 days, as was evident in the nerve crush model. **Conclusion:** Temporary injury and vocal fold immobilization result in transient shifts in MHC composition. Nerve transection and repair result in

persistent alteration of MHC composition and vocal fold dysfunction. The expression of normal MHC composition is dependent on the condition of appropriate neural contact and functional reinnervation. **Key Words:** Myosin heavy chain, nerve injury, cricoarytenoid joint fixation, rat larynx, recurrent laryngeal nerve.

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INTRODUCTION

Neural,^{1–3} hormonal,^{4,5} and mechanical^{6–8} factors influence and maintain postmaturational expression of myosin heavy chain (MHC) composition. Activity-dependent expression of MHC composition in rat laryngeal muscles has been demonstrated under conditions of normal laryngeal development⁹ and following laryngeal denervation in adult animals.¹⁰ In normally developing animals, the appearance of adult MHC phenotype follows onset of functional activity (i.e., swallowing, respiration) and reflects the physiological character of each muscle. In the denervated condition, observed changes in MHC composition in laryngeal muscle include an increase in type IIA + IIX, a decrease in type IIB, a variable change in type IIL, and a minimal change in type I MHC. Although reinnervation of muscle may alter the MHC phenotype of denervated muscle, muscle reinnervation demonstrated by electromyography (EMG) rarely achieves normal functional activity. This condition, generally referred to as synkinesis, implies incomplete or inappropriate reinnervation.

In the present study, the MHC composition of laryngeal muscle is assessed using sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) under varying degrees of nerve injury to determine whether MHC expression is influenced by the states of innervation and vocal fold motion. Nerve crush (CR) injury, nerve transection with repair, and joint fixation (JF) are used to simulate clinically important conditions resulting in vocal fold motion impairment: recurrent laryngeal nerve (RLN) traction injury, RLN transection and repair during surgery, and cricoarytenoid joint ankylosis. If expression correlates with degree of recovery, MHC composition may then be used as a biological marker for functional reinnervation.

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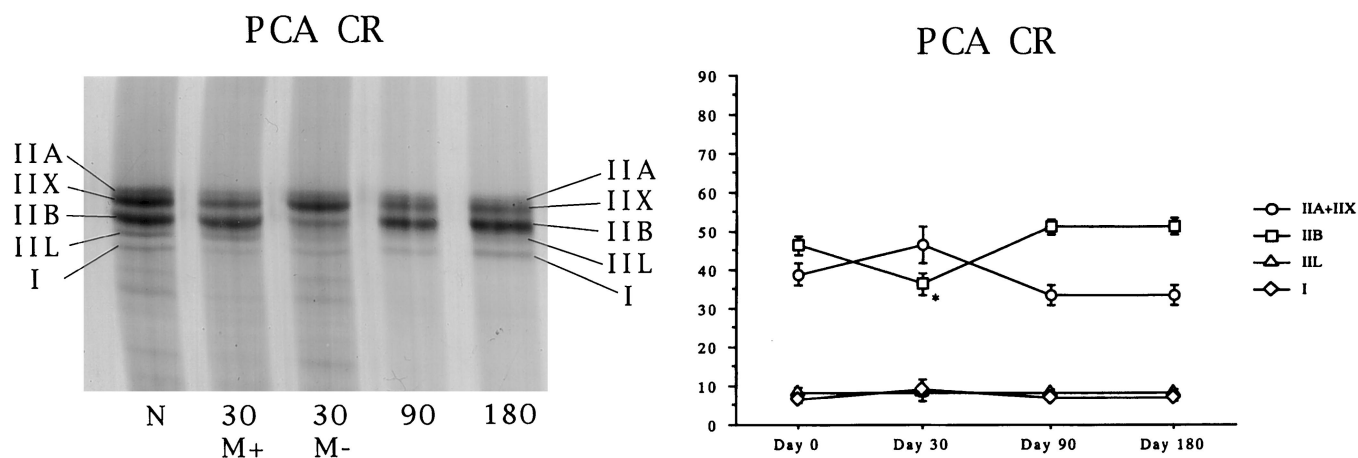


Fig. 1. Change in relative myosin heavy chain composition after nerve crush in the posterior cricoarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE ($n = 6$). Representative sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) gel is shown (left). M+ = animals with vocal fold motion; M- = animals with impaired vocal fold motion.

MATERIALS AND METHODS

Surgery and Tissue Preparation

Institutional guidelines in accordance with the Animal Welfare Act of 1966 and all subsequent revisions including those made in 1985 and National Institutes of Health guidelines were followed for the handling and care of laboratory animals.

Fifty-four Sprague-Dawley rats (weight, 200–250 g) were anesthetized by intramuscular injection of ketamine hydrochloride 35 mg/kg and xylazine hydrochloride 5 mg/kg. Animals were divided into three groups representing the three experimental conditions, nerve crush CR ($n = 18$), nerve transection and repair (TR) ($n = 18$), and cricoarytenoid JF ($n = 18$).

By means of a standard operating microscope, the larynx was approached through a midline incision. In the CR group, the left RLN was carefully separated from the tracheoesophageal groove. A 1-mm-long crush injury was produced by applying a constant force for 60 seconds with smooth-tip forceps. In the TR group, the left RLN was divided sharply and immediately repaired end-to-end with 10-0 nylon microsuture. In the cricoarytenoid JF group, a midline thyrotomy was performed to visualize the laryngeal cavity. The left cricoarytenoid joint capsule was

opened, and the articulating surface macerated. The larynx was closed using 10-0 nylon suture. After each procedure, left vocal fold motion impairment was confirmed using a laryngeal telescope and the skin was closed.

Six animals from each group were allowed to survive for 30, 90, and 180 days. At each time interval, animals were anesthetized as described above, and vocal fold motion was assessed by direct visualization with a laryngeal telescope. In the CR and TR groups, a midline thyrotomy was performed before animals were killed and laryngeal EMG performed to confirm reinnervation. Laryngeal EMG was performed using a 27 G bipolar EMG needle and GW Instruments (Elkins-Sinn Inc., Cherry Hill, NJ) analogue-digital converter and software. Animals were euthanized using a lethal dose of sodium pentobarbital. The posterior cricoarytenoid (PCA) and thyroarytenoid (TA) were rapidly microdissected from each animal.

Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis

Laryngeal muscle tissue from individual animals was processed separately. Sample preparation was performed as de-

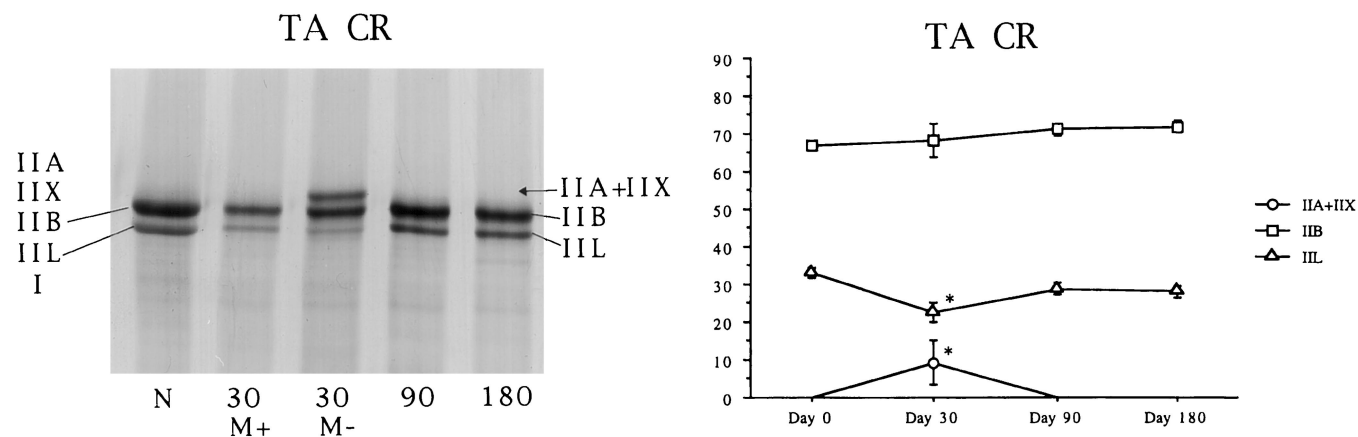


Fig. 2. Change in relative myosin heavy chain composition after nerve crush in the thyroarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE for the denervated condition ($n = 6$). Representative SDS-PAGE gel is shown (left).

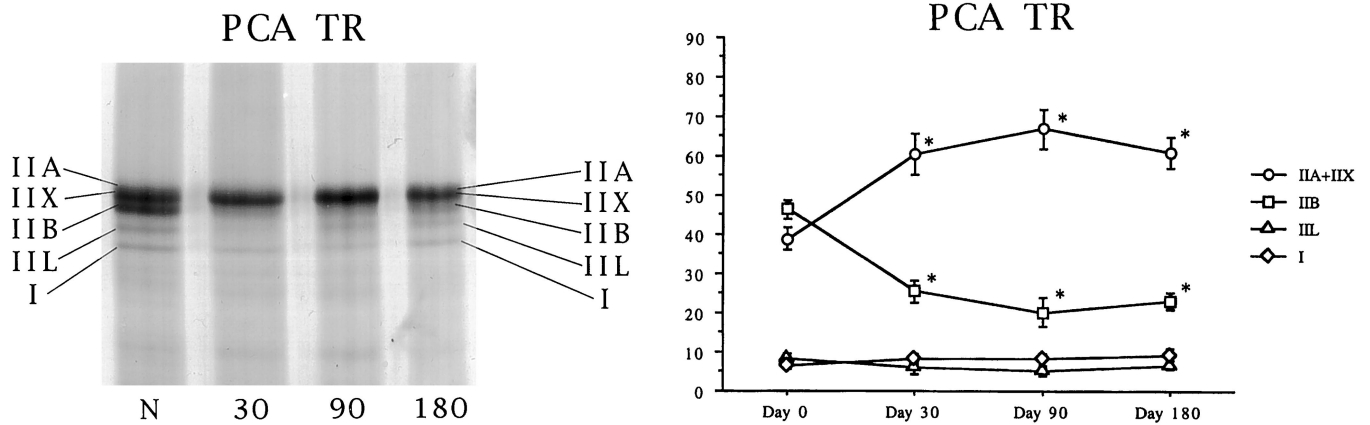


Fig. 3. Change in relative myosin heavy chain composition after nerve transection and repair in the posterior cricoarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE for the denervated condition ($n = 6$). Representative SDS-PAGE gel is shown (left).

scribed previously.^{9–12} SDS-PAGE was performed using a 0.75-mm-thick 5.5% acrylamide–35% glycerol separating gel and a 3% acrylamide stacking gel.^{9–11} The gels were developed with silver stain to visualize the protein bands, using Silver Stain Plus Kit (Biorad, Hercules, CA). Individual stained gels were digitally recorded and analyzed densitometrically using NIH image 1.58f software (Scientific Computing Resource Center, Bethesda, MD).

Data Analysis

The relative MHC composition values in six individual animals at each stage were obtained from the silver-stained gels using optical densitometry. One-way ANOVA followed by Fisher's protected least significant difference test was performed using Statview SE version 1.04 software (Abacus Concepts Inc., Berkeley, CA) to compare changes in percentage of MHC expressed for each survival period. Results are reported as mean $\pm$ standard error. Statistical significance is set at $P < .05$.

RESULTS

Vocal Fold Motion and Electromyography

Nerve crush group. Normal vocal fold motion was observed in three of six animals at 30 days following CR.

No motion was detected in the remaining three animals. Normal vocal fold motion was observed in 6 of 6 animals at 90 and at 180 days following CR. In the three animals with motion impairment, evidence of reinnervation was present with polyphasic motor unit potentials and 1 to 3+ recruitment with inspiration and swallow (range, 0–4+).

Nerve transection and repair group. Complete vocal fold immobility was observed in six of six animals at 30, 90, and 180 days. Electromyography demonstrated evidence of reinnervation in all animals. In general, polyphasic motor unit potentials and 3+ recruitment with respiration and swallow were observed.

Joint fixation group. Vocal fold immobility was observed in six of six animals at 30, 90, and 180 days. Laryngeal electromyography was not performed in this group.

Nerve Crush Group: Myosin Heavy Chain Compositional Change

Posterior cricoarytenoid. The MHC changes in posterior cricoarytenoid (PCA) (Fig. 1) that were observed following CR are demonstrated in a representative SDS-

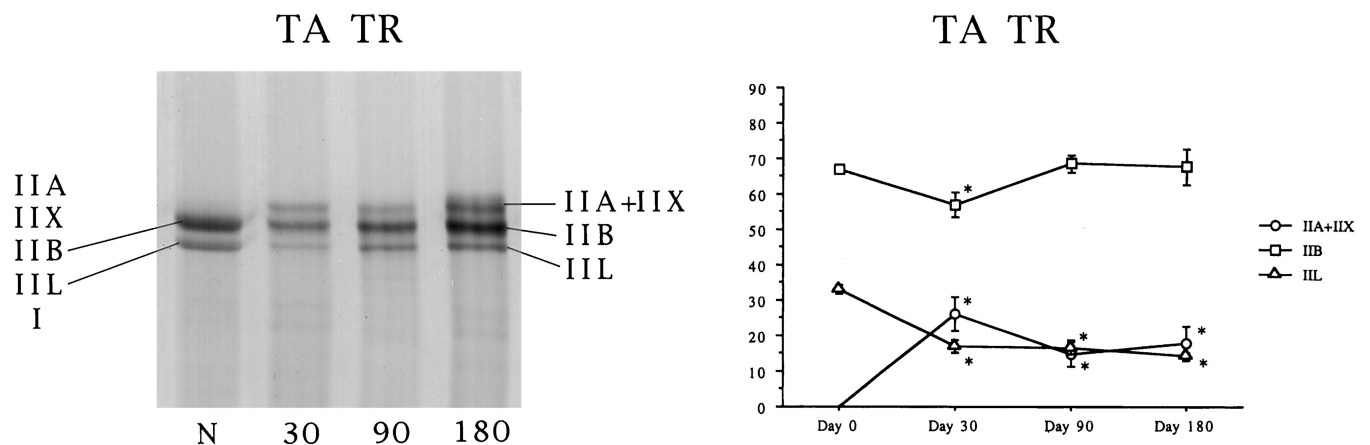


Fig. 4. Change in relative myosin heavy chain composition after nerve transection and repair in the thyroarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE for the denervated condition ($n = 6$). Representative SDS-PAGE gel is shown (left).

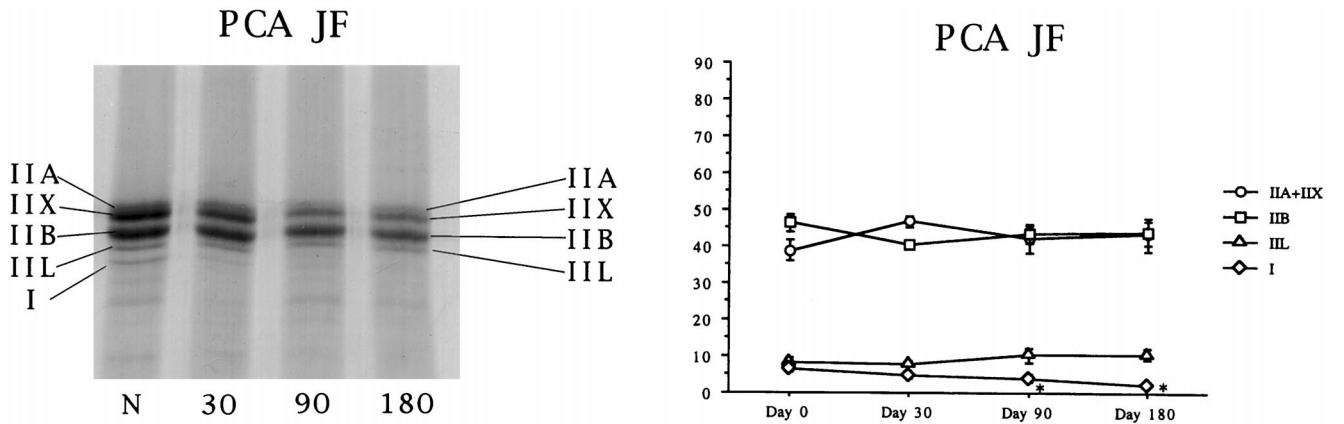


Fig. 5. Change in relative myosin heavy chain composition after cricoarytenoid joint fixation in the posterior cricoarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE ($n = 6$). Representative SDS-PAGE gel is shown (left).

PAGE gel. The mean value ($n = 6$) for relative percentage of composition of each MHC band is demonstrated with day 0 representing the normal condition. A significant change in MHC composition was observed for the type IIB component 30 days after CR injury. Animals with vocal fold motion (M+) at day 30 were observed to have a normal MHC profile, whereas animals with impaired vocal fold motion (M-) were found to have MHC profiles that are characteristic of denervated muscle. Although a corresponding increase in type IIA+IIX MHC occurred at day 30, the change was not significant. Types I and IIL MHC remained stable following CR, and normal values was observed on days 90 and 180.

Thyroarytenoid. The MHC changes in TA (Fig. 2) that were observed following CR are demonstrated in a representative SDS-PAGE gel. The mean value ($n = 6$) for relative percentage of composition of each MHC band is demonstrated with day 0 representing the normal condition. In the normal condition, TA is composed of predominately type IIB and a smaller percentage of type IIL. Type IIA+X appears transiently on day 30, and a corresponding significant decrease in type IIL is observed. Type IIB MHC remains unchanged following crush injury, and all

MHC components normalize on days 90 and 180. As observed in the PCA muscle, animals with M+ at day 30 were observed to have a normal MHC profile, whereas animals with M- were found to have MHC profiles that are characteristic of denervated muscle.

Nerve Transection and Repair Group: Myosin Heavy Chain Compositional Change

Posterior cricoarytenoid. A representative gel for each time interval is shown (Fig. 3). A significant decrease in type IIB and a significant increase in type IIA+IIX (characteristic of denervated muscle) were observed in all animals 30, 90, and 180 days following TR. No significant change was observed in the type I or type IIL MHC bands following TR.

Thyroarytenoid. A representative gel and graphic display of MHC compositional change are shown in Figure 4. There is a significant decrease in both type IIB and type IIL on day 30 with the appearance of type IIA+X (characteristic of denervated muscle). The type IIA+X component persists on days 90 and 180, and type IIL remains significantly diminished. Type IIB normalizes after day 30.

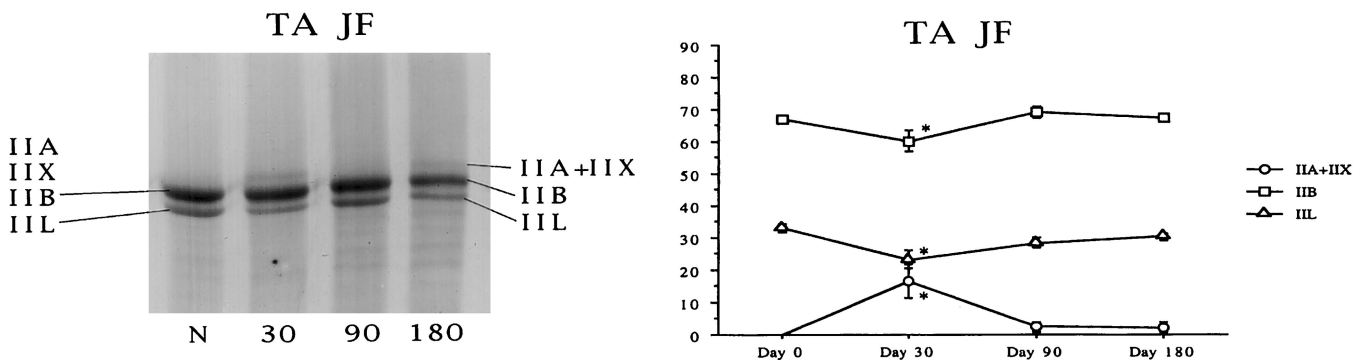


Fig. 6. Change in relative myosin heavy chain composition after cricoarytenoid joint fixation in the thyroarytenoid muscle. Day 0 represents the normal condition and is reported as mean value $\pm$ SE ($n = 6$). Subsequent days represent the mean value $\pm$ SE ($n = 6$). Representative SDS-PAGE gel is shown (left).

Cricothyroid Joint Fixation Group: Myosin Heavy Chain Compositional Change

Posterior cricothyroid. A representative gel and graphic display of MHC compositional change in the PCA following JF are shown (Fig. 5). A small but significant decrease in type I MHC was observed 90 and 180 days following cricothyroid JF. No significant change in types IIA+IIX, IIB, or IIL was observed.

Thyroarytenoid. A representative gel and graphic display of MHC compositional change in the TA following JF are shown (Fig. 6). The appearance of type IIA+X MHC and corresponding significant decrease in types IIB and IIL were observed on day 30 following JF. Normalization of all MHC types is seen on days 90 and 180.

DISCUSSION

Postmaturational expression of the adult MHC phenotype in mammalian muscle is influenced by multiple factors, including neural activity, mechanical stretch, and both systemic and local trophic factors.^{1–8} Disruption of neural input to the soleus hindlimb muscle in rats is characterized by a decrease in type I MHC, an increase in type IIA MHC, and the appearance of types IIB and IIX MHC.^{13,14} Similar findings have been reported in laryngeal muscle where, in the denervated condition, changes in MHC composition include an increase in type IIA+IIX, a decrease in type IIB, variable change in type IIL, and minimal change in type I MHC.¹⁰ Alterations in MHC phenotype may also occur in conditions of muscle inactivity with intact neural connections, as observed in hindlimb suspension studies.⁶ In non-weight-bearing conditions it is assumed that the absence of mechanical stretch reduces the expression of local trophic factors, thus affecting change in MHC expression.^{6–8}

In the present study, MHC composition is used as a biological marker in varying conditions of laryngeal motion impairment. Nerve crush injury, TR, and cricothyroid JF are used to simulate clinically important conditions resulting in vocal fold motion impairment: RLN traction injury, RLN transection and repair during surgery, and cricothyroid joint ankylosis. In the crush injury model, denervated muscles are reinnervated by their original motoneuron pool and full recovery of function is anticipated.^{15,16} In the transection and repair model, complete division of all axons is performed and reinnervation of muscle occurs in a random manner^{17,18} resulting in synkinesis and persistent motion impairment.

Following crush injury, transient alteration in MHC composition was associated with impaired vocal fold function. The MHC change observed is characteristic of denervated muscle and was seen only in animals with motion impairment and EMG findings that were consistent with denervation. Animals with normal motion following crush injury displayed normal MHC patterns. In contrast, TR with reinnervation by EMG resulted in persistent vocal fold motion impairment and MHC alterations that were consistent with denervation. These findings are indicative of laryngeal synkinesis or inappropriate reinnervation and imply that reinnervation alone is insufficient to influence MHC expression.

Comparable changes in MHC expression have been reported in hindlimb muscle following CR and nerve transection.¹³ Following sciatic nerve crush injury, transient increase in type IIA/IIX MHC and a decrease in type IIB MHC were observed in the tibialis anterior muscle at 8 weeks. Functional recovery with normal MHC expression was seen at 32 weeks. In contrast, nerve transection resulted in persistent alteration of MHC expression at 8 and 32 weeks.¹³

Cricothyroid JF results in a condition similar to mechanical stretch models used in hindlimb studies, where neural integrity is preserved and muscle contraction is isometric.^{7,8,19,20} With hindlimb JF, muscle may be fixed in a stretched (loaded) position or shortened (unloaded) position.⁷ Muscle loading produces an increase in type I MHC and a decrease in type IIX.⁸ Unloaded muscle has been characterized by a decrease in type I MHC and an increase in type IIX.^{19,20}

Cricothyroid JF produced an early change in MHC expression in the TA only, similar to that observed following CR, with a decrease in types IIB and IIL and appearance of type IIA/IIX MHC. A delayed decrease in percentage of type I MHC was observed in the PCA muscle only, at 90 and 180 days. The early change in TA muscle MHC may represent RLN trauma similar to that observed in the CR group. As an alternative, a transient change in MHC expression may reflect a reduction in muscle activity and corresponding decrease in muscle loading, comparable to hindlimb unloading. The decrease in type I MHC observed in the PCA muscle would also be consistent with unloading and is comparable to the MHC changes observed in unloaded hindlimb muscle.

CONCLUSION

The findings in this study indicate that neural input alone is not sufficient to support normal MHC expression. Moreover, activity-dependent MHC expression requires functional recovery as determined by normal muscle contraction. Based on these observations, normalization of MHC expression is indicative of appropriate and functional reinnervation.

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